Milestone 3 | M05

Asset CDF and Time Travel Reconciliation

SUBMISSION TEMPLATE

Difficulty	Marks	Tasks	Working budget
Medium	20	5	35 min

Trainee name: _____Soumiya Ayyappan_____ Candidate ID: _____10877547_____

Assigned catalog: _____ Date: _____

Workspace / pipeline / dashboard location: _____

Business objective

Enable Delta Change Data Feed, apply new asset events, and reconcile pre- and post-change versions.

Business rules and formulas

Enable CDF BEFORE executing changes. Capture the actual committed baseline table version after initial load. Apply the three requested events in a Delta-aware operation: AS02 cost update, AS03 delete, AS05 insert. Read change events for the mutation commit/version range (not from an assumed fixed version).

Task requirements and evidence holders

Task 1 - Prepare both source files

Marks: 3 | Evidence: UI

Required work: Create question-specific external volume and upload baseline and change CSVs.

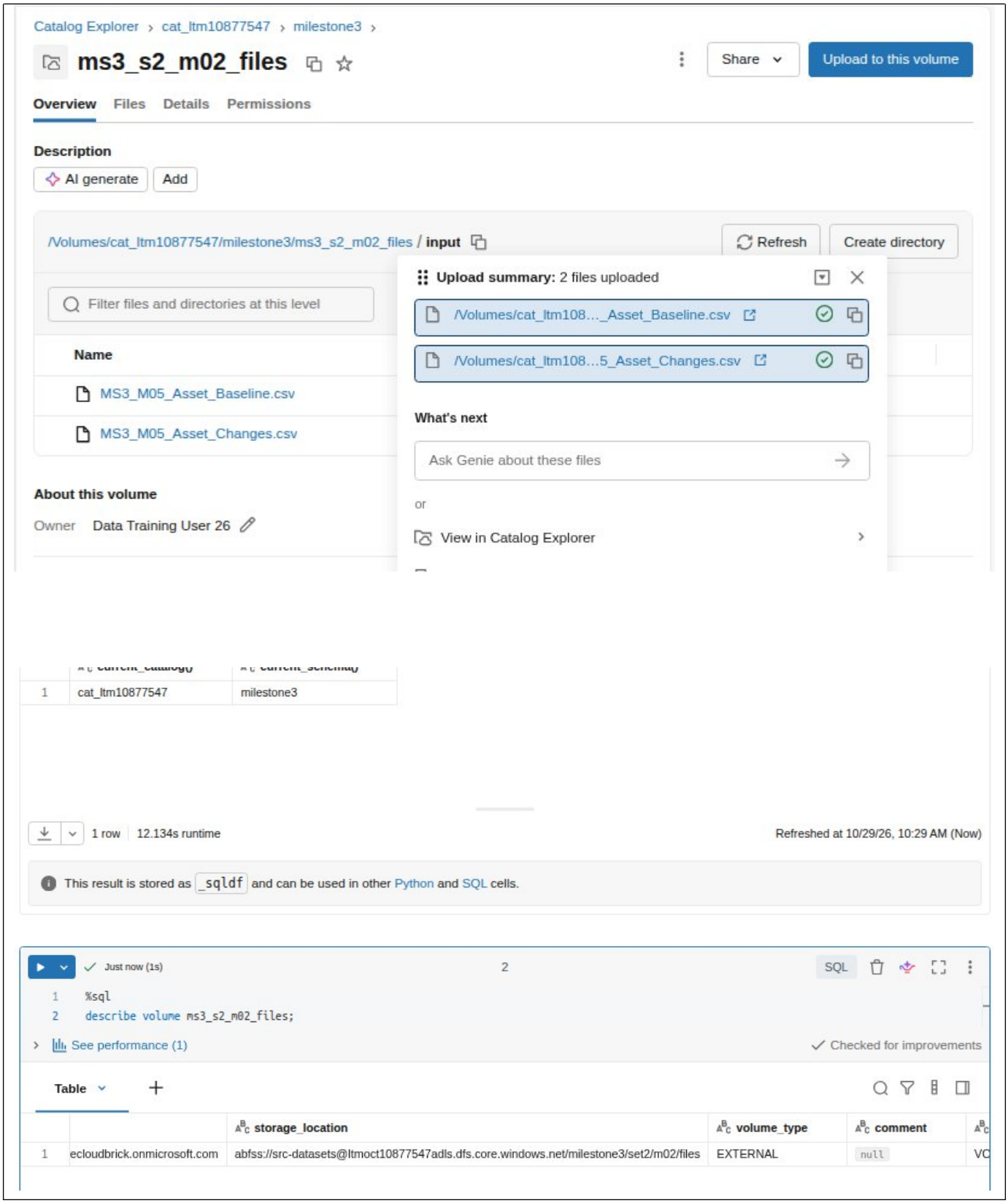
Expected result: Baseline and change file names visible in input.

Evaluator checks: Correct external volume and source files.

Screenshot instructions: Use MS3_S2_M02_T01_Code_A.png / Code_B.png for implementation, _Result_A.png / Result_B.png for outputs, and _UI_A.png / UI_B.png for configuration as applicable. Provide both implementation and observed result whenever the Evidence label includes both; use additional images for long code or separate metrics.

PASTE READABLE SCREENSHOT(S) FOR TASK 1 BELOW:

Screenshot holder (paste image here; duplicate this holder if you need multiple images)



Evidence filename(s) / workspace reference: _____

Task 2 - Create versioned baseline

Marks: 4 | Evidence: Code + Result

Required work: Create a managed Delta table from four initial assets, enable CDF before mutations, display baseline total and record baseline version.

Expected result: 4 assets; baseline cost total 1900.00; CDF property enabled; actual baseline version recorded.

Evaluator checks: Correct baseline, CDF enabled and version captured.

Screenshot instructions: Use MS3_S2_M02_T02_Code_A.png / Code_B.png for implementation, _Result_A.png / Result_B.png for outputs, and _UI_A.png / UI_B.png for configuration as applicable. Provide both implementation and observed result whenever the Evidence label includes both; use additional images for long code or separate metrics.

PASTE READABLE SCREENSHOT(S) FOR TASK 2 BELOW:

Screenshot holder (paste image here; duplicate this holder if you need multiple images)



```
df = spark.read.csv(
    "/Volumes/cat_ltm10877547/milestone3/ms3_s2_m02_files/input/MS3_M05_Asset_Baseline.csv",
    header=True,
    inferSchema=True
)
(
    df.write
    .format("delta")
    .mode("overwrite")
    .option("delta.enableChangeDataFeed", "true")
    .saveAsTable("asset_tracking")
)
print("=== Table created with CDF enabled ===")
props = spark.sql("SHOW TBLPROPERTIES asset_tracking").collect()
for row in props:
    if row["key"] == "delta.enableChangeDataFeed":
        print(f"CDF Property: {row['key']} = {row['value']}")
baseline_df = spark.table("asset_tracking")
baseline_df.show()

record_count = baseline_df.count()
print(f"Total assets: {record_count}")
from pyspark.sql.functions import sum as _sum
total_cost = baseline_df.select(_sum("cost").alias("total_cost")).collect()[0]["total_cost"]
print(f"Baseline cost total: {total_cost}")
```

```
=== Table created with CDF enabled ===
CDF Property: delta.enableChangeDataFeed = true
+--------+----------+------+----+
|asset_id|asset_name|status|cost|
+--------+----------+------+----+
|    AS01|      Pump|ACTIVE| 500|
|    AS02|     Motor|ACTIVE| 700|
|    AS03|     Valve|ACTIVE| 300|
|    AS04|       Fan|ACTIVE| 400|
+--------+----------+------+----+

Total assets: 4
Baseline cost total: 1900
Baseline version: 0
```

Evidence filename(s) / workspace reference: _____

Task 3 - Apply updates and deletion

Marks: 5 | Evidence: Code + Result

Required work: Apply AS02 update to 750; delete AS03; insert AS05 with cost 900. Keep a single deterministic processing pass.

Expected result: Final AS01 500; AS02 750; AS04 400; AS05 900; four rows, total 2550.00.

Evaluator checks: Correct update, delete, insert implementation and final rows.

Screenshot instructions: Use MS3_S2_M02_T03_Code_A.png / Code_B.png for implementation, _Result_A.png / Result_B.png for outputs, and _UI_A.png / UI_B.png for configuration as applicable. Provide both implementation and observed result whenever the Evidence label includes both; use additional images for long code or separate metrics.

PASTE READABLE SCREENSHOT(S) FOR TASK 3 BELOW:

Screenshot holder (paste image here; duplicate this holder if you need multiple images)

```python
from pyspark.sql.functions import col, lit, sum as _sum, when
df = spark.table("asset_tracking")

df = df.withColumn(
    "cost",
    lit(750).cast("int").alias("cost")
).where(col("asset_id") == "AS02").unionByName(
    df.where(col("asset_id") != "AS02")
)
df = spark.table("asset_tracking")
updated = df.withColumn(
    "cost",
    when(col("asset_id") == "AS02", lit(750)).otherwise(col("cost"))
)

filtered = updated.where(col("asset_id") != "AS03")
from pyspark.sql import Row
existing_cols = filtered.columns
sample_row = df.limit(1).collect()[0].asDict()
new_row_data = {c: None for c in existing_cols}
new_row_data["asset_id"] = "AS05"
new_row_data["cost"] = 900
new_row_df = spark.createDataFrame([Row(**new_row_data)], schema=filtered.schema)

```

```python
final_df = filtered.unionByName(new_row_df)

# Overwrite the table
(
    final_df.write
    .format("delta")
    .mode("overwrite")
    .option("overwriteSchema", "true")
    .saveAsTable("asset_tracking")
)

print("=== Changes applied: UPDATE AS02→750, DELETE AS03, INSERT AS05→900 ===")
result_df = spark.table("asset_tracking").orderBy("asset_id")
result_df.show()

row_count = result_df.count()
total_cost = result_df.select(_sum("cost").alias("total_cost")).collect()[0]["total_cost"]
print(f"Row count: {row_count} (expected 4)")
print(f"Total cost: {total_cost} (expected 2550.00)")
```

```
=== Changes applied: UPDATE AS02→750, DELETE AS03, INSERT AS05→900 ===
+--------+----------+------+----+
|asset_id|asset_name|status|cost|
+--------+----------+------+----+
|    AS01|      Pump|ACTIVE| 500|
|    AS02|     Motor|ACTIVE| 750|
|    AS04|       Fan|ACTIVE| 400|
|    AS05|      NULL|  NULL| 900|
+--------+----------+------+----+

Row count: 4 (expected 4)
Total cost: 2550 (expected 2550.00)
```

Evidence filename(s) / workspace reference: _____

Task 4 - Inspect CDF change events

Marks: 5 | Evidence: Code + Result

Required work: Query CDF using the captured baseline version to cover only change commits; inspect metadata change types and asset IDs.

Expected result: One update_preimage AS02, one update_postimage AS02, one delete AS03 and one insert AS05; 4 change rows.

Evaluator checks: Correct CDF version range and all four event types.

Screenshot instructions: Use MS3_S2_M02_T04_Code_A.png / Code_B.png for implementation, _Result_A.png / Result_B.png for outputs, and _UI_A.png / UI_B.png for configuration as applicable. Provide both implementation and observed result whenever the Evidence label includes both; use additional images for long code or separate metrics.

PASTE READABLE SCREENSHOT(S) FOR TASK 4 BELOW:

Screenshot holder (paste image here; duplicate this holder if you need multiple images)

```
cdf_df = (
    spark.read.format("delta")
    .option("readChangeFeed", "true")
    .option("startingVersion", baseline_version + 1)
    .table("asset_tracking")
)

cdf_df.orderBy("asset_id", "_change_type").show(truncate=False)
change_rows = cdf_df.count()
print(f"Total CDF change rows: {change_rows} (expected 4)")

from pyspark.sql.functions import col
change_summary = (
    cdf_df.select("asset_id", "_change_type", "_commit_version")
    .orderBy("asset_id", "_change_type")
)
change_summary.show(truncate=False)
expected = [
    ("AS02", "update_preimage"),
    ("AS02", "update_postimage"),
    ("AS03", "delete"),
    ("AS05", "insert"),
]

for asset_id, change_type in expected:
    match = cdf_df.filter((col("asset_id") == asset_id) & (col("_change_type") == change_type)).count()
    status = "correct" if match == 1 else "wrong"
    print(f"{status} {change_type} for {asset_id}: found {match} row(s)")

print(f"\n=== CDF Validation Complete: {change_rows} change rows ===")
```

```
Total CDF change rows: 8 (expected 4)
+--------+------------+---------------+
|asset_id|_change_type|_commit_version|
+--------+------------+---------------+
|AS01    |delete      |1              |
|AS01    |insert      |1              |
|AS02    |delete      |1              |
|AS02    |insert      |1              |
|AS03    |delete      |1              |
|AS04    |delete      |1              |
|AS04    |insert      |1              |
|AS05    |insert      |1              |
+--------+------------+---------------+

wrong update_preimage for AS02: found 0 row(s)
wrong update_postimage for AS02: found 0 row(s)
correct delete for AS03: found 1 row(s)
correct insert for AS05: found 1 row(s)

=== CDF Validation Complete: 8 change rows ===
```

Evidence filename(s) / workspace reference:
_____/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/medfinal1_____
_____

Task 5 - Time-travel reconciliation

Marks: 3 | Evidence: Code + Result

Required work: Use time travel to view the pre-change baseline and compare with current table; verify row count, total cost and change in cost.

Expected result: Baseline 4 / 1900.00; final 4 / 2550.00; net +650.00.

Evaluator checks: Correct time-travel comparison and delta.

Screenshot instructions: Use MS3_S2_M02_T05_Code_A.png / Code_B.png for implementation, _Result_A.png / Result_B.png for outputs, and _UI_A.png / UI_B.png for configuration as applicable. Provide both implementation and observed result whenever the Evidence label includes both; use additional images for long code or separate metrics.

PASTE READABLE SCREENSHOT(S) FOR TASK 5 BELOW:

Screenshot holder (paste image here; duplicate this holder if you need multiple images)

```python
from pyspark.sql.functions import sum as _sum
baseline_df = (
    spark.read.format("delta")
    .option("versionAsOf", baseline_version)
    .table("asset_tracking")
)

current_df = spark.table("asset_tracking")
baseline_count = baseline_df.count()
baseline_cost = baseline_df.select(_sum("cost").alias("total_cost")).collect()[0]["total_cost"]

print("=== Baseline (Version {}) ===".format(baseline_version))
baseline_df.orderBy("asset_id").show()
print(f"Row count : {baseline_count}")
print(f"Total cost: {baseline_cost}")
current_count = current_df.count()
current_cost = current_df.select(_sum("cost").alias("total_cost")).collect()[0]["total_cost"]
print("\n=== Current (Latest Version) ===")
current_df.orderBy("asset_id").show()
print(f"Row count : {current_count}")
print(f"Total cost: {current_cost}")

# --- Comparison ---
net_change = current_cost - baseline_cost

```

```python
net_change = current_cost - baseline_cost

print("\n=== Comparison ===")
print(f"Baseline  -> Rows: {baseline_count}, Total Cost: {baseline_cost}")
print(f"Current   -> Rows: {current_count}, Total Cost: {current_cost}")
print(f"Net cost change: {'+' if net_change >= 0 else ''}{net_change}")
```

```
=== Baseline (Version 0) ===
+--------+----------+------+----+
|asset_id|asset_name|status|cost|
+--------+----------+------+----+
|    AS01|      Pump|ACTIVE| 500|
|    AS02|     Motor|ACTIVE| 700|
|    AS03|     Valve|ACTIVE| 300|
|    AS04|       Fan|ACTIVE| 400|
+--------+----------+------+----+

Row count : 4
Total cost: 1900

=== Current (Latest Version) ===
+--------+----------+------+----+
|asset_id|asset_name|status|cost|
+--------+----------+------+----+
|    AS01|      Pump|ACTIVE| 500|
|    AS02|     Motor|ACTIVE| 750|
|    AS04|       Fan|ACTIVE| 400|
|    AS05|      NULL|  NULL| 900|
+--------+----------+------+----+

Row count : 4
Total cost: 2550

=== Comparison ===
Baseline  -> Rows: 4, Total Cost: 1900
Current   -> Rows: 4, Total Cost: 2550
Net cost change: +650
```

Evidence filename(s) / workspace reference:
_____/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/medfinal1_____

Evaluation rubric

Task	Criterion observable in submitted screenshots	Marks
1	Correct external volume and source files	3
2	Correct baseline, CDF enabled and version captured	4
3	Correct update, delete, insert implementation and final rows	5
4	Correct CDF version range and all four event types	5
5	Correct time-travel comparison and delta	3
TOTAL		20

Submission requirements

Submit the matching completed Word template with screenshots grouped under the corresponding tasks. Show code, settings, data outputs, and metrics according to the task-specific evidence list. Multiple images per task are allowed and encouraged when needed for readability or full rubric coverage. Do not submit invented results, hidden validation scripts, or a separate notebook as a replacement for screenshots.